

Wall-resolved SA flat-plate diagnostic

Re_L=100000, production SA/SIMPLE, 64x56 stretched wall-resolved grid.

Actual 1800 iterations; no equilibrium wall function in SA transport.

Original 20% regression limit is not treated as 3% qualification.

skin_friction_relative_error: 0.0722281087; limit=0.03; pass=False

solver_residual: 6.91502141e-05; limit=1e-05; pass=False

convergence_failure: 1; limit=0.5; pass=False

maximum_local_yplus: 2.09015512; limit=1; pass=False

Independent wall force Cf=0.006865512; reference=0.0074.

Pressure has zero x-force on a flat wall.

The fully turbulent empirical correlation is not exact DNS/experiment.

Current SA and momentum residuals fail the steady gate.

The 128x112 CPU/CUDA comparison covers five production iterations only.

Next: residual convergence, matched public data, grid refinement,

then consistent wall-function turbulence boundaries.

